

Budget Justification (max. 2 pages)

The budget requested from SSHRC is \$499,540.

Personnel costs (\$464,000)

The largest portion of the costs are allocated towards the students (master's and PhD) and post-doctoral research fellows. Stipends for master's students are \$25,000 per year (years 1-2; 3-4), stipends of PhD students are \$28,000 per year (years 1-4; 2-5), and the Post-Doctoral Fellows will be paid \$70,000 per year (years 1 & 5). Over the duration of this 5-year project, this will account for 92.8% of the funds requested. A timeline of when master's students, PhD students, and post-doctoral fellows will be working on the project, and the tasks allocated to them, can be seen in **Table T1**.

Budget Item	Funds Requested
Student stipends	
Master's Student Stipends (4 student-years x \$25,000)	\$100,000
PhD Student Stipends (8 student-years x \$28,000)	\$224,000
Non-student salaries	
Post-Doctoral Salaries (2 person-years x \$70,000)	\$140,000
Total	\$464,000

Travel costs (\$21,000)

Beginning in year 2 of the project, each student will receive \$800 per year for Canadian travel to attend domestic conferences and meeting such as the Canadian Associations of Geographers Annual Conference or transportation-oriented conferences such as the Canadian Transportation Research Forum. This translates to \$2,400 in years 2-4 of the project, and \$800 in year 5, when the only student will be PhD2. Additionally, starting on year 2 of the project, each student will receive \$1,200 for international travel to conferences such as the Annual Meeting of the Transportation Research Board or NECTAR. This translates to \$3,900 per year in years 2 to 4 of the project and \$1,300 in year 5, when only PhD2 will be in the project. Regardless of year of study, students will be encouraged to attend the conferences as well as present in their upper years of study, as it is crucial for not only showcasing the research project, but also learning from other experts in the fields of transportation, geography, and planning. Hence, 4.2% of the total project cost will be allocated to travel costs.

Budget Item	Funds Requested
Travel and subsistence	
Students: Canadian travel (10 student-years x \$800)	\$8,000
Students: Foreign travel (10 student-years x \$1,300)	\$13,000
Total	\$21,000

Computer hardware (\$7,500)

The quantitative research segments of this project, namely the geospatial analysis, the development of the equity-informed evaluation tool, and the development of open-source packages, will require specialized equipment. This translates into computers with sufficient processors, RAM, and storage to be able to complete various computing processes and generating data. Members of the research team have some computers in their equipment inventory, and these will be used for this project. Funds are requested to upgrade or replace ageing equipment on years 2 and 3 of the project. We are budgeting \$3,500 in year 2 of the project, and \$3,000 in year 3 of the project to this end. In addition, we are

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requesting \$500 in each of years 4 and 5. These funds will be allocated to hardware updates, computer repairs, and the acquisition of various peripheral equipment as needed. Together with computer equipment already owned by members of the research team, the aim is to maintain four fully equipped workstations for the duration of the project, with one workstation assigned to each of the work packages.

Budget Item	Funds Requested
Non-disposable equipment	
Acquire computer equipment/replace ageing equipment (Year 2)	\$3,000
Acquire computer equipment/replace ageing equipment (Year 3)	\$3,000
Repair/peripherals (year 4)	\$500
Repair/peripherals (year 5)	\$500
Total	\$7,000

Other expenses: Codesign (\$7,540)

The qualitative research component of the project is the CODESIGN, with an estimated budget of \$6,000 to provide incentives for participants across different focus groups (based on factors such as socioeconomic status and/or neighbourhood). Each codesign focus group participant will receive a \$50 gift card and their transport to and from McMaster reimbursed ($\$4.00 \times 2$ (return bus fare) = \$8.00). All co-design focus group materials (e.g., printing, Bristol boards, etc.) will be provided by Ravensbergen's lab in kind. There will be two codesign phases: development of indicators and development of tools.

Budget Item	Funds Requested
Other: Codesign (Development of indicators)	
Step 1 (Empathy) Four focus groups with 5 stakeholders (Year 2) ($4 \times 5 \times \$58$)	\$1,160
Steps 2 and 3 (Define and Ideate) Three focus groups with 5 transit users and 2 focus groups with 5 transit providers (Year 2) ($3 \times 5 \times \$58 + 2 \times 5 \times \58)	\$1,450
Step 4 (Prototype) Done in-house by the research team (year 3)	\$0
Step 5 (Test) Three focus groups with 4 transit users and two focus groups with 4 transit providers (year 3) ($3 \times 4 \times \$58 + 2 \times 4 \times \58)	\$1,160
Other: Codesign (Development of tools)	
Step 1 (Empathy) Four focus groups with 5 stakeholders (Year 4) ($4 \times 5 \times \$58$)	\$1,160
Steps 2 and 3 (Define and Ideate) Three focus groups with 5 transit users and 2 focus groups with 5 transit providers (Year 4) ($3 \times 5 \times \$58 + 2 \times 5 \times \58)	\$1,450
Step 4 (Prototype) Done in-house by the research team (year 5)	\$0
Step 5 (Test) Three focus groups with 4 transit users and two focus groups with 4 transit providers (year 5) ($3 \times 4 \times \$58 + 2 \times 4 \times \58)	\$1,160
Total	\$7,540

Software (\$0)

This project will exclusively use open-source software.

Open access publishing (\$0)

We will leverage publishing agreements made by our institutions with reputable publishers such as Wiley, PLoS ONE, Elsevier, and Springer.